Table 6.2: Unsaturated fatty acids

Symbol	Structure	Common name
16:1 (Δ^9)	$CH_3(CH_2)_5\ CH{=}CH(CH_2)_7\ COOH$	Palmitoleic acid
18:1 (Δ^9)	$CH_3(CH_2)_7\ CH{=}CH(CH_2)_7\ COOH$	Oleic acid
18:1 (Δ^9) (trans)	$CH_3(CH_2)_7\ CH{=}CH(CH_2)_7\ COOH$ (trans)	Elaidic acid
Essential fatty acids		
18:2 ($\Delta^{9,12}$)	$CH_3(CH_2)_4\ CH{=}CHCH_2CH{=}CH(CH_2)_7COOH$	Linoleic acid
18:3 ($\Delta^{9,12,15}$)	$CH_3CH_2CH{=}CHCH_2CH{=}CHCH_2CH{=}CH(CH_2)_7COOH$	Linolenic acid
20:4 ($\Delta^{5,8,11,14}$)	$CH_3(CH_2)_4\ (CH{=}CHCH_2)_3CH{=}CH(CH_2)_3COOH$	Arachidonic acid

Essential fatty acid (EFA)

Fatty acids that must be obtained from the diet as they cannot be synthesized by our body are essential fatty acids (Figure 6.5). EFA are polyunsaturated fatty acids. They are precursors of prostaglandins, a family of physiologically potent lipids.

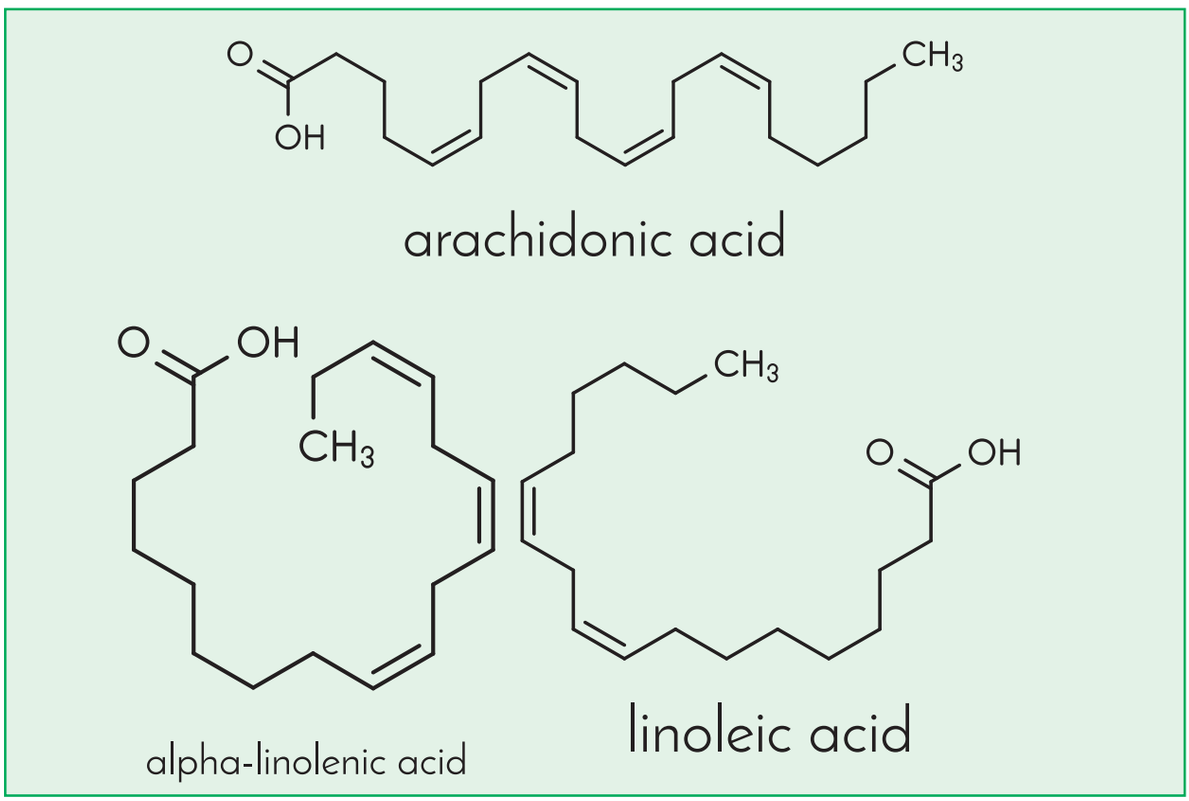


Figure 6.5 Essential fatty acids

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